Clustering 2

Introduction to Data Mining

Seoul National University

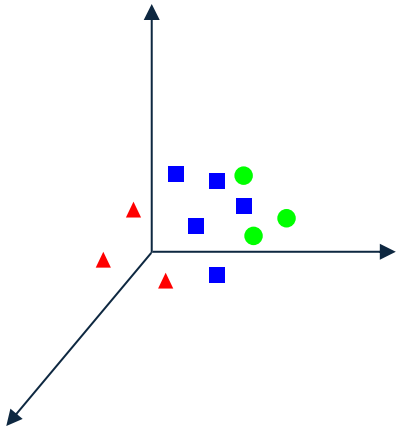
Spring 2026

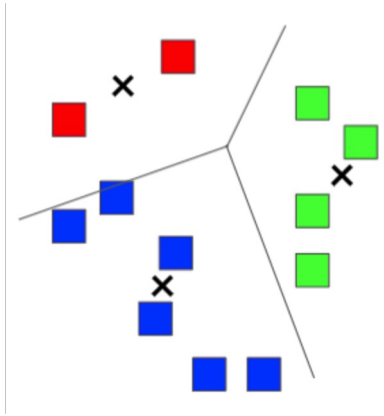
Announcements

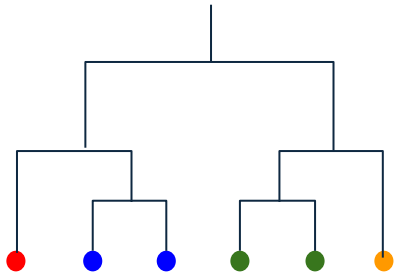
- **Assignments:**
  - HW1 is posted (deadline: April 9).

Recap

- Clustering
  - Curse of dimensionality
- Hierarchical clustering
  - Euclidean vs. non-Euclidean
  - Centroids vs. clustroids
- k-means clustering
  - k-means++
  - Selection of k







Outline

1. **<u>CURE Algorithm</u>**
2. BFR Algorithm
3. BFR Algorithm: Process
4. GRGPF Algorithm

Overview

- **Q:** How to do clustering if the data is too large to fit in memory?
- **CURE:** Simplest approach based on hierarchical clustering.
- **BFR:** Variant of k-means for large data.
  - More advanced but introduces more assumptions compared to CURE.
- **GRGPF:** Clever approach for a non-Euclidean space.
  - Summarizes a cluster without its centroid.

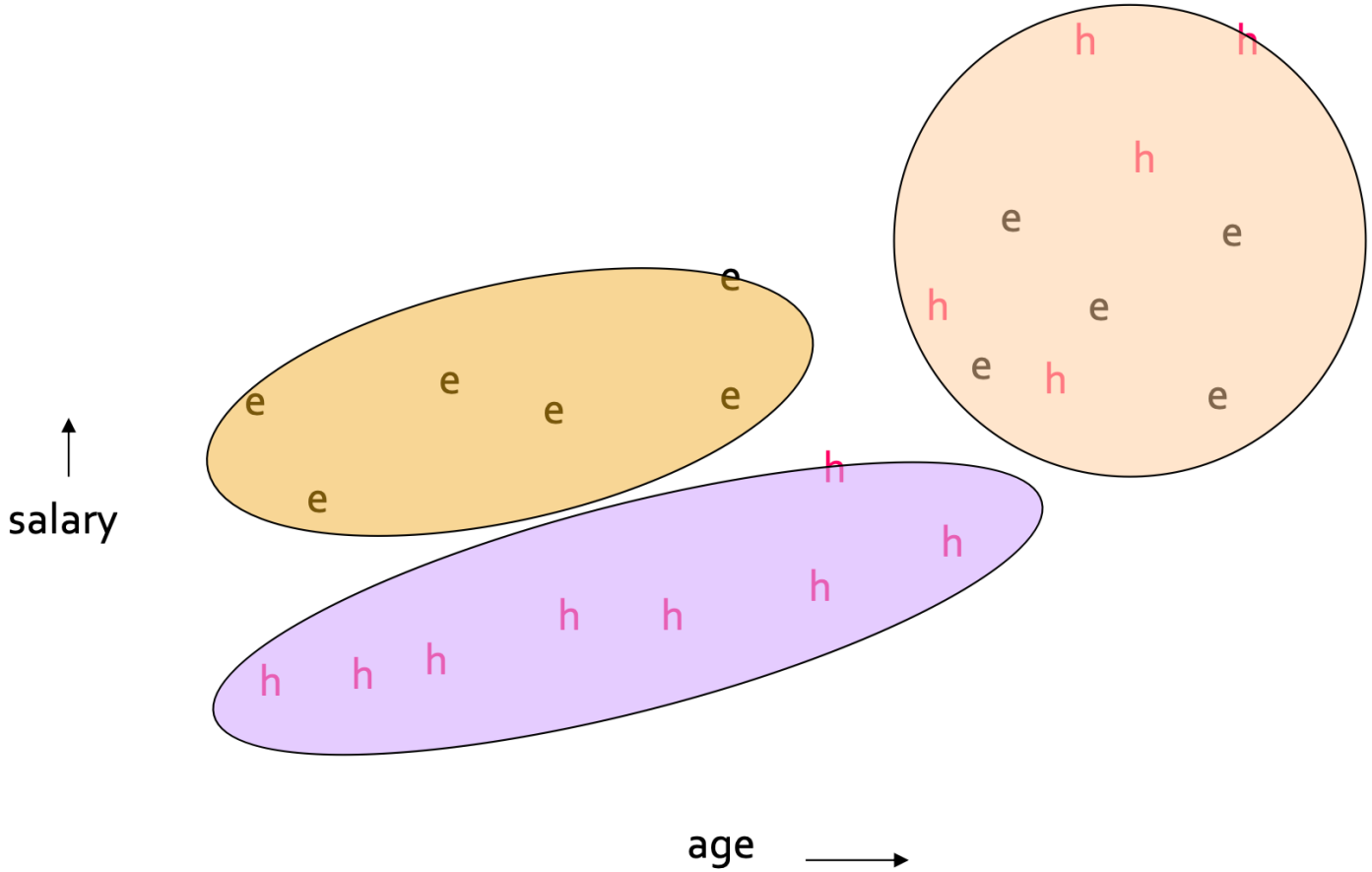
CURE (Clustering Using REpresentatives)

- **CURE** is a 2-pass algorithm for large disk-resident data.
  - **Pass 1:** Find clusters from a sampled subset.
  - **Pass 2:** Assign points to the clusters.
- Use a collection of representative points to represent clusters.
- Assume Euclidean space/distance, with k (# of clusters) given.

CURE: Pass 1

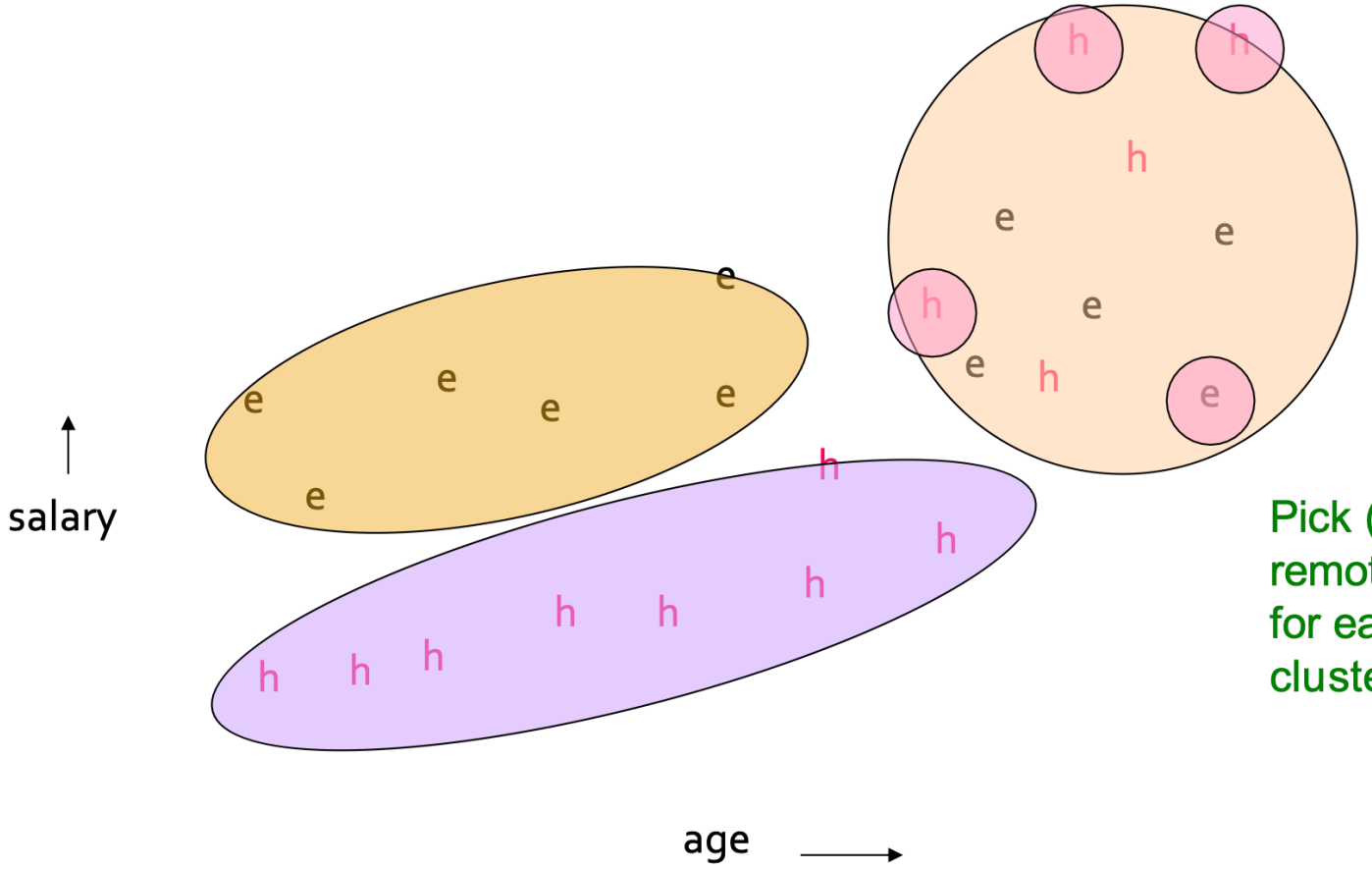
1. Pick a random sample of data.
2. Cluster them in main memory using hierarchical clustering.
   - Merge two clusters when they have close pairs of points.
3. Pick representative points from each cluster.
   - For each cluster, pick a sample of points, as dispersed as possible.
   - Move them a fraction of distance, e.g. 20%, toward the centroid … why?
4. Merge clusters with the closest pair of representatives.

Example: Picking Dispersed Points



Source: Stanford CS246 (2022)

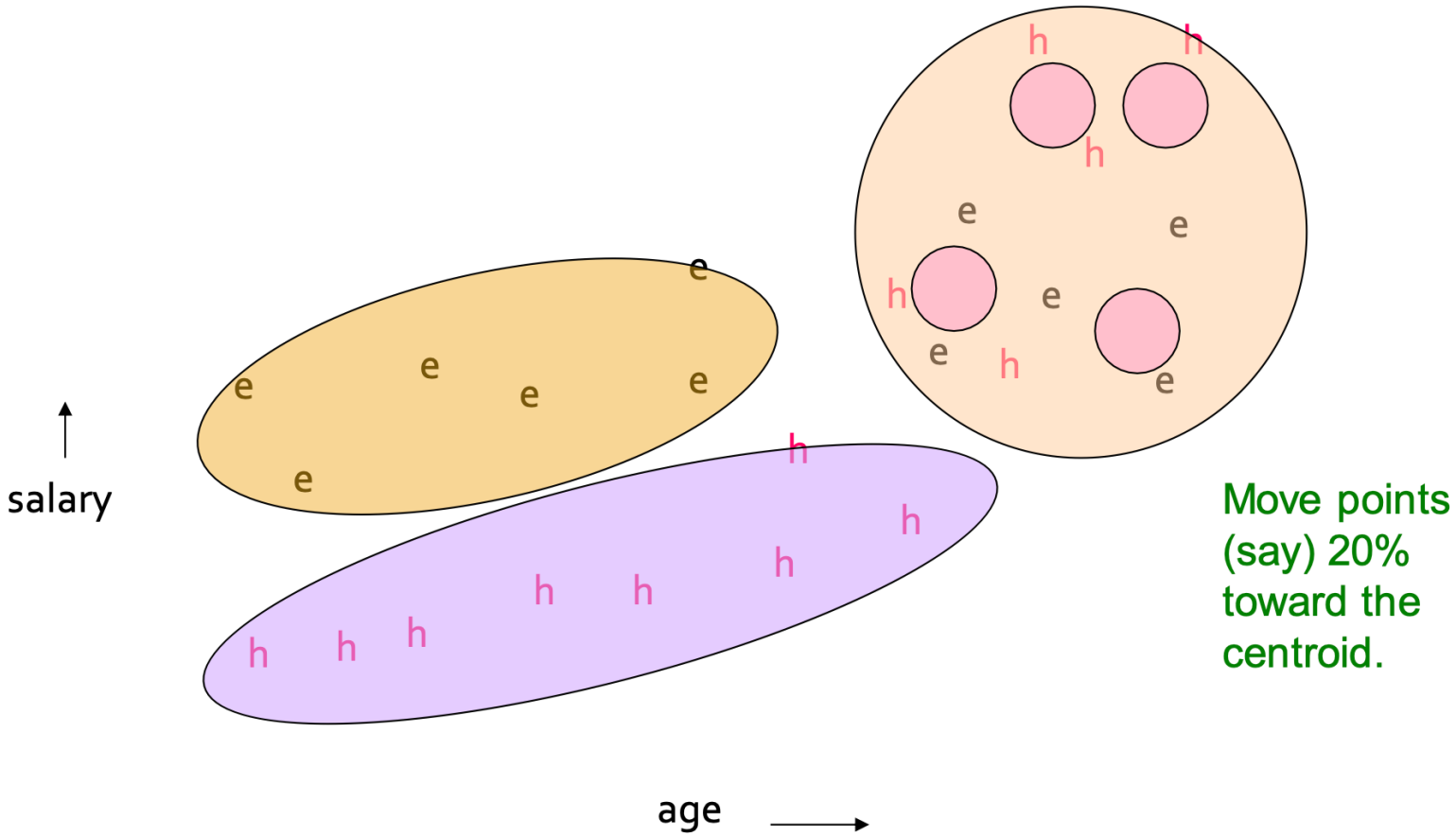
Example: Picking Dispersed Points



Pick (say) 4 remote points for each cluster.

Source: Stanford CS246 (2022)

Example: Picking Dispersed Points



Source: Stanford CS246 (2022)

CURE: Pass 2

1. Rescan the whole dataset.
2. Assign each point p in the dataset to the "closest cluster."
   - Find the closest representative point to p.
   - Assign p to the representative's cluster.

Why to Move 20% Inward?

- Some of the representatives will be on the boundary of clusters.
- We move them 20% towards the centroids.
- Large, dispersed clusters will shrink more than small, dense ones
- As a result, the algorithm favors a small, dense cluster.

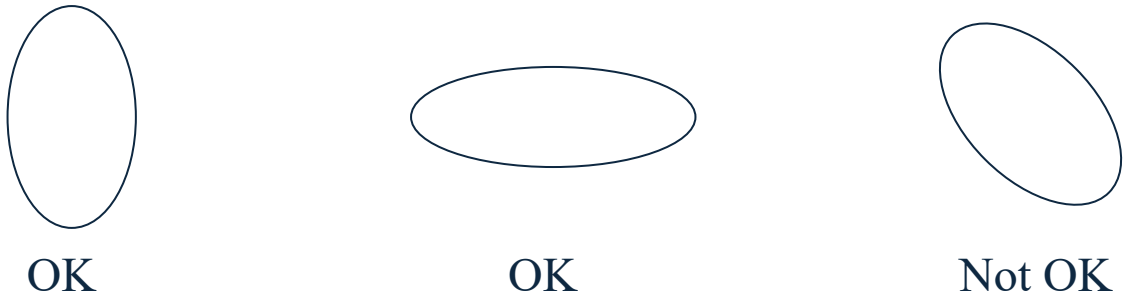
Source: Stanford CS246 (2022)

Outline

1. CURE Algorithm
2. **<u>BFR Algorithm</u>**
3. BFR Algorithm: Process
4. GRGPF Algorithm

BFR (Bradley, Fayyad, and Reina) Algorithm

- Variant of k-means for large disk-resident data.
- Assume normally-distributed clusters in a Euclidean space.
  - Clusters are axis-aligned ellipses.
  - Standard deviations in different dimensions may vary.
- **Goal:** Find cluster centroids efficiently (in Pass 1).
  - Point assignment can be done in Pass 2 through whole data.



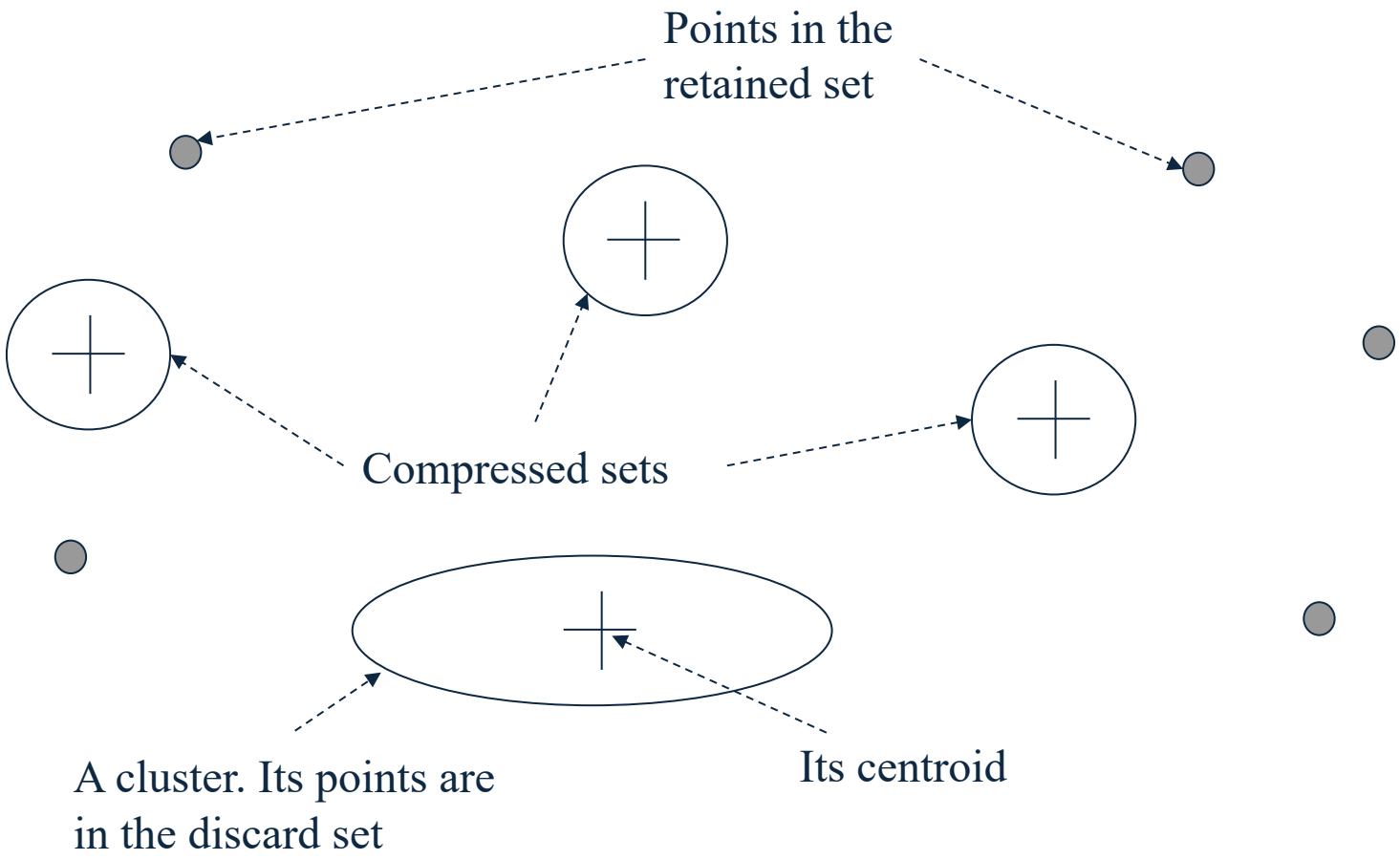
Main Idea

- **Idea:** Keep summary statistics of groups of points.
  1. Read points from disk one main-memory-full at a time.
  2. Summarize most points from previous memory loads.
  3. Memory requirement changes from $O(\text{data})$ to $O(\text{clusters})$.
- **Challenge:** Not all points can be summarized effectively.
  - Some points will not create a cluster at all.

Three Classes of Points

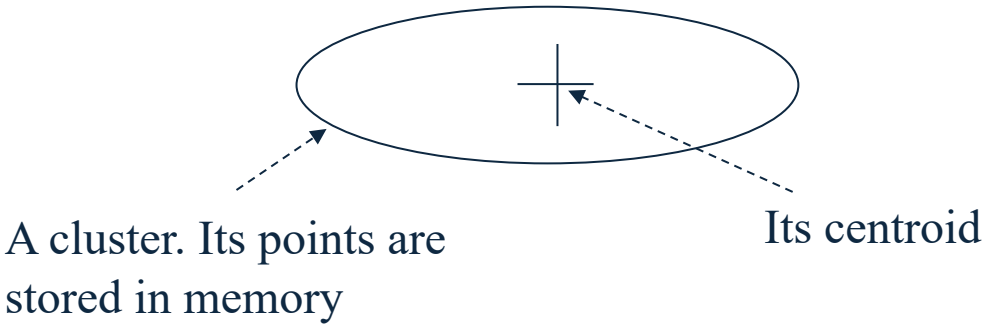
- Divide all data points into three sets during the algorithm.
- **Discard set** (= clusters):
  - Points close to a centroid to be **summarized** and discarded.
- **Compressed set** (= miniclusters):
  - Groups of points that are close together but not to any centroid.
  - These points are **summarized** but not assigned to a cluster.
- **Retained set:**
  - Isolated points waiting to be assigned to a compressed set.

Cluster Visualization



Summarizing Sets of Points

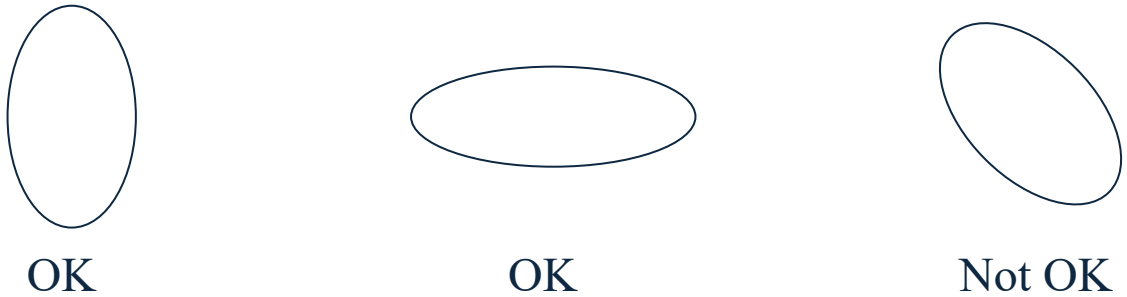
- Discard or compressed set is summarized by $2d + 1$ values.
  - The number N of points.
  - A vector SUM, where SUM_i is the sum of the coordinates of all points.
  - A vector SUMSQ, where SUMSQ_i is the sum of squared coordinates.
    - $\mathrm{SUM}_i = \sum_{k \in C} x_{ki}$ and $\mathrm{SUMSQ}_i = \sum_{k \in C} x_{ki}^2$.



C: Set of points
d: # of dimensions

Summarizing Sets of Points

- We can compute the centroid and the size of each cluster.
  - **Average** in dimension i (= centroid): SUM_i/N.
  - **Variance** in dimension i (= cluster size): $\mathrm{SUMSQ}_i/N - (\mathrm{SUM}_i/N)^2$.
    - Because $\mathrm{Var}(X) = E[X^2] - E[X]^2$.
  - **Standard deviation** is the square root of variance.
- This is based on the assumption of "axis-aligned" clusters.
  - Without it, we need to store the $d \times d$ covariance matrix.



Benefits of the Representation

- **Benefit 1:** Easy to add a new point to a cluster.
  - Increase N by 1.
  - Add the vector to SUM.
  - Add the squares of components to SUMSQ.
- **Benefit 2:** Easy to combine two clusters.
  - Add corresponding values of N, SUM, and SUMSQ.

Pop Quiz

- Represent the cluster of points $(5, 1)$, $(6, -2)$, and $(7, 0)$.
  - N = ?
  - SUM = ?
  - SUMSQ = ?
- Compute the variance of the first dimension.
  - Variance = ?

Outline

1. CURE Algorithm
2. BFR Algorithm
3. **<u>BFR Algorithm: Process</u>**
4. GRGPF Algorithm

Overview of the BFR Algorithm

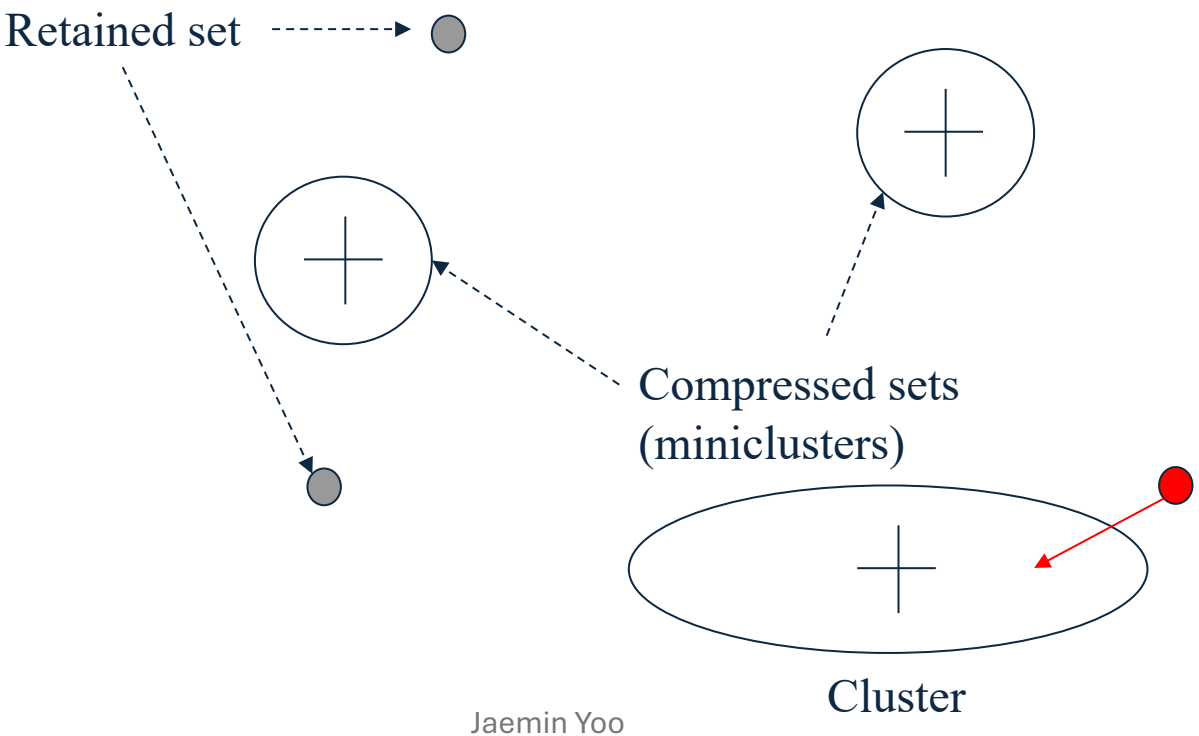
1. Initialize k clusters/centroids (as in k-means)
2. **for** each chunk **in** a data file
3. **for** each point **in** the chunk
4. Assign it to a cluster/minicluster if it is *sufficiently close* to it
5. Cluster the remaining points, creating new *miniclusters*
6. Try to merge existing clusters and miniclusters if they are close

Selection of the Initial Centroids

- The k initial centroids can be selected as in k-means:
- **Option 1:** Take k random points.
- **Option 2:** Take a small sample, cluster it, and use the centroids.
- **Option 3:** Take a sample, pick a random point, and $k - 1$ more.
  - Each as far from the previously selected points as possible.

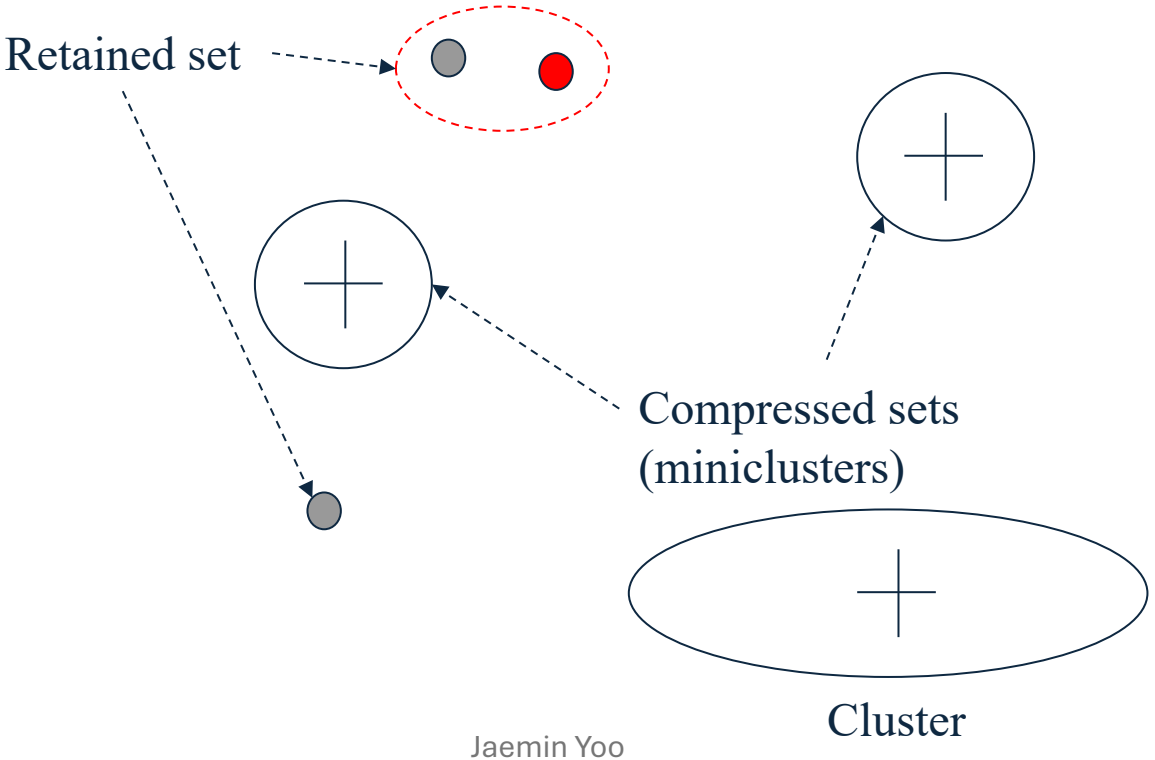
Processing a Chunk of Points (1/5)

- All points that are “sufficiently close” to any centroid:
  - Added to that cluster or minicluster.



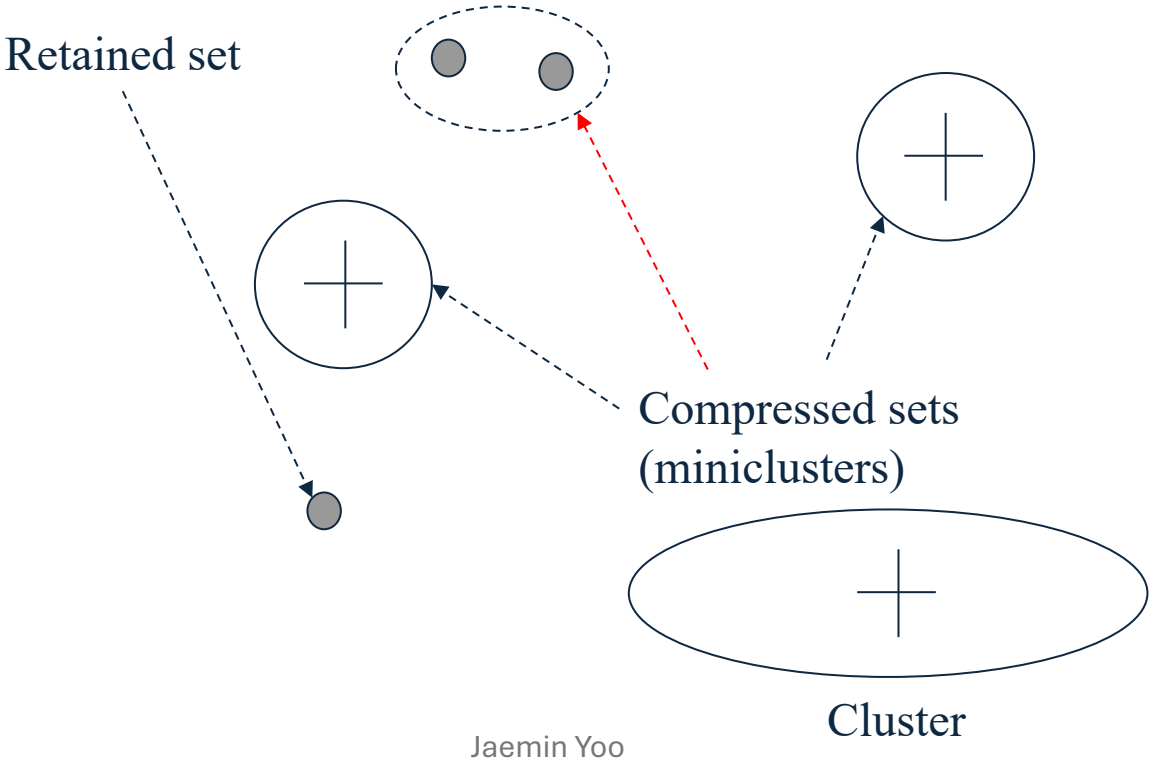
Processing a Chunk of Points (2/5)

- All points that are not sufficiently close to any centroid:
  - Clustered with the points in the retained set using any algorithm.



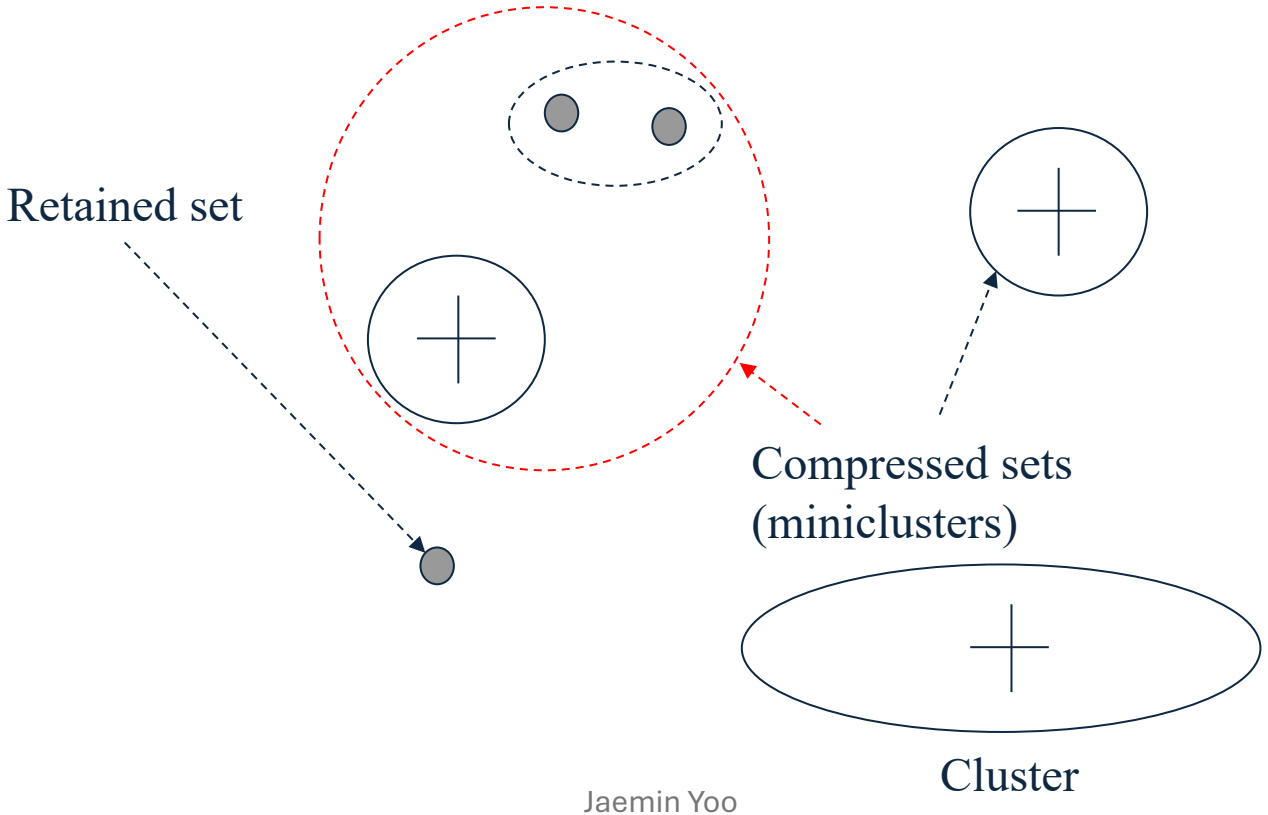
Processing a Chunk of Points (3/5)

- New groups of ≥ 2 points become miniclusters.
- Singleton groups remain in the retained set.



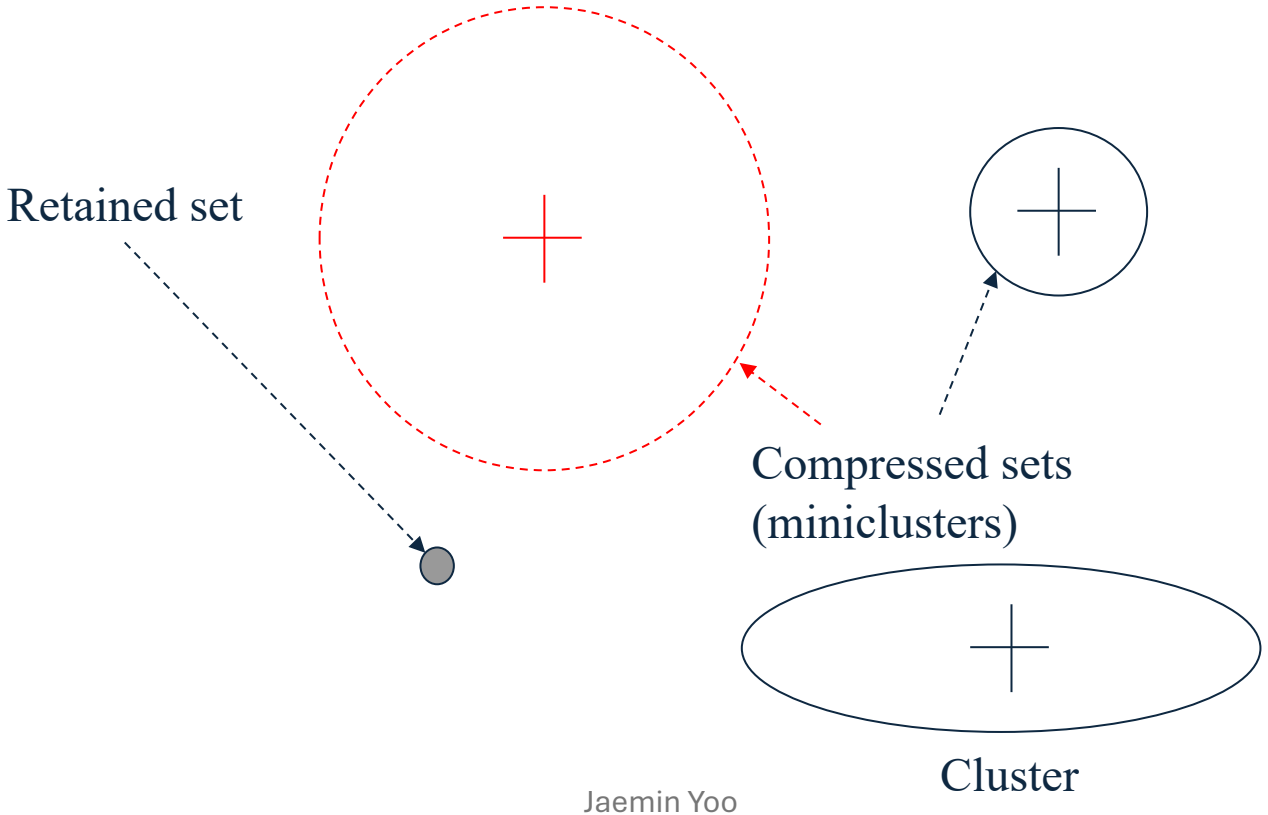
Processing a Chunk of Points (4/5)

- Combine the clusters and miniclusters if they are close enough.



Processing a Chunk of Points (5/5)

- Points assigned to a cluster or a minicluster are written to disk.



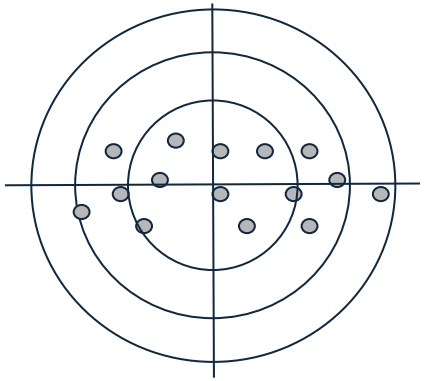
After Processing All Chunks

- Finally, what to do with compressed and retained sets?
- **Option 1:** Treat them as outliers and never cluster them.
- **Option 2:** Assign each of them to the nearest cluster.
  - For both individual points and miniclusters.

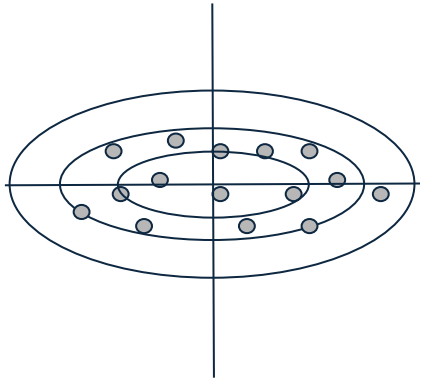
How Close is Close Enough?

- Need a way to decide whether to put a new point into a cluster.
- BFR compares the **Mahalanobis distance** with a threshold.
  - **Definition:** $\sqrt{\sum_{i=1}^{d}\left(\frac{p_i - c_i}{\sigma_i}\right)^2}$
  - Euclidean distance from the centroid c normalized by standard dev. σ_i.
  - Leverages the assumption that points are normally distributed.

Euclidean distance:

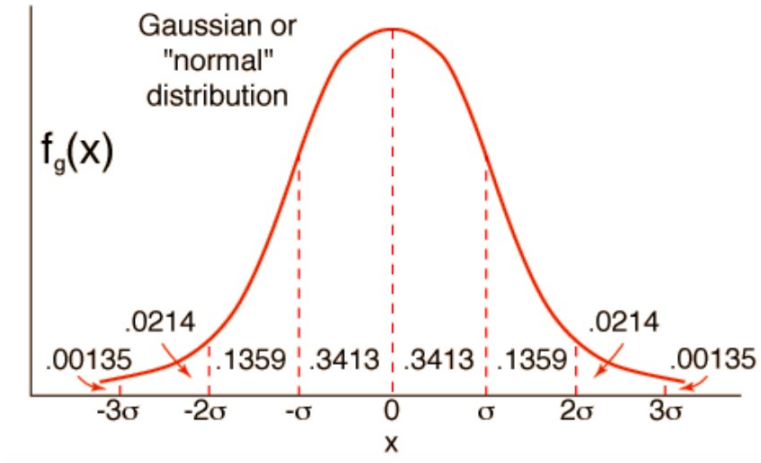


Mahalanobis distance:



Assigning a Point to a Cluster

- For each point p, choose the cluster with the closest centroid.
- Add p to the cluster if the distance is less than a threshold.
  - Suppose the threshold $= 4$.
  - Then, $\Pr(\text{value being 4 standard deviations from the mean}) < 10^{-6}$.



Source: Stanford CS246 (2022)

When to Merge Two Clusters?

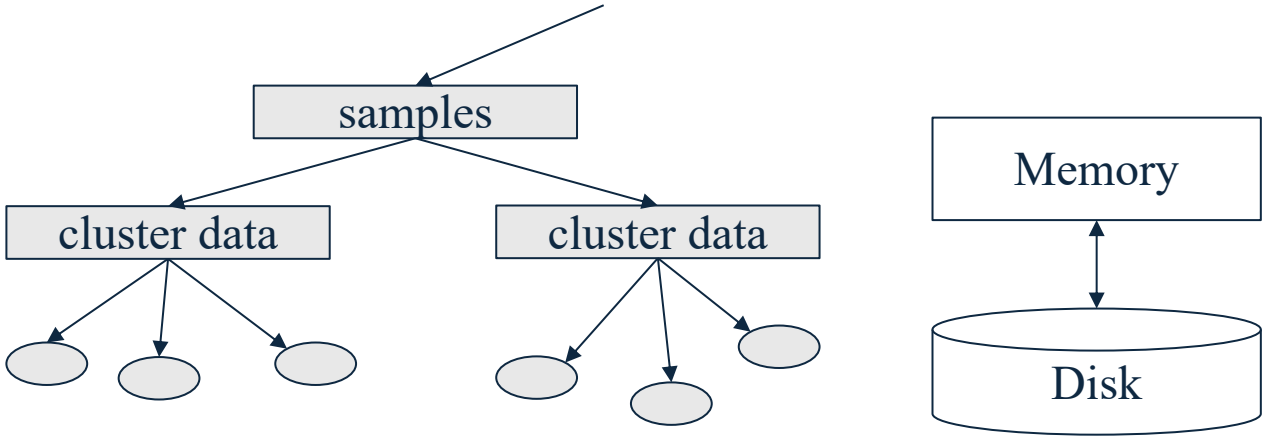
- Compute the variance of the combined subcluster.
- Combine if the combined variance is below some threshold.
- **Many alternatives:** E.g., considering density.

Outline

1. CURE Algorithm
2. BFR Algorithm
3. BFR Algorithm: Process
4. **<u>GRGPF Algorithm</u>**

GRGPF Algorithm

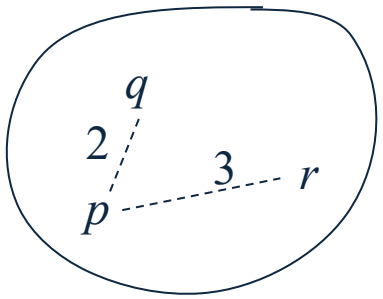
- Does not require a Euclidean space unlike CURE and BFR.
- Represents clusters by **well-chosen sample points** in memory.
- Organizes clusters hierarchically, as a tree (not covered today).
  - New point is assigned to a cluster by passing it down the tree.



Cluster Representation

- How to represent (or summarize) a cluster in GRGPF:
  - The number N of points.
  - The clustroid c.
  - The **rowsum** of the clustroid.
    - Sum of the squares of the distances from p to each point in the cluster.
  - k points that are **nearest** to the clustroid, and their rowsums.
  - k points that are **farthest** from the clustroid, and their rowsums.

$$\textit{rowsum} \text{ of point } p \text{ in cluster } C = \sum_{c \in C} d(p, c)^2$$

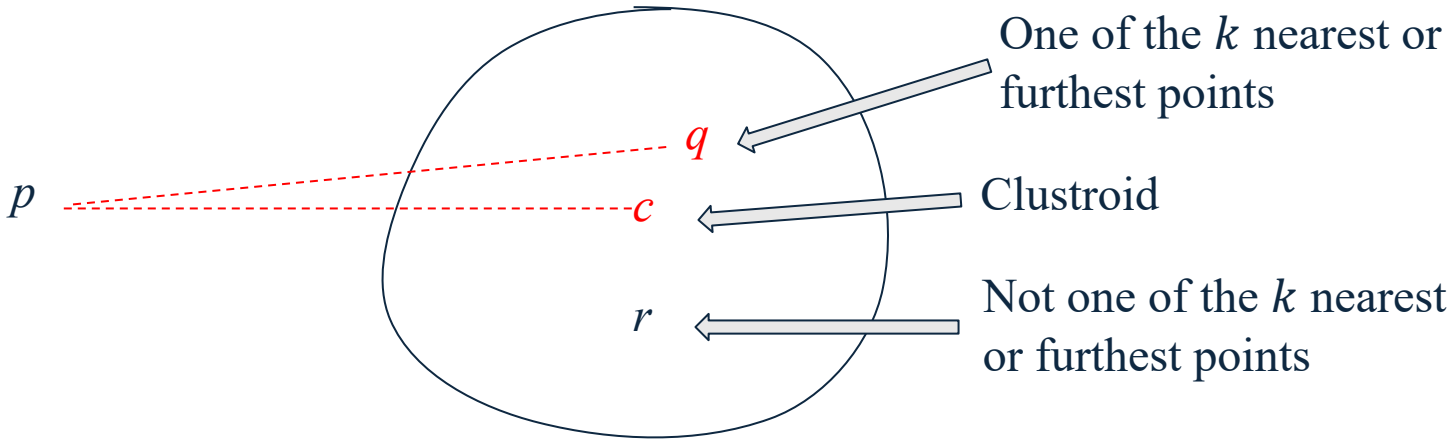


Justification of the Representation

- Clustroid is the point in the cluster with the smallest rowsum.
- Why the k points nearest to the clustroid?
  - If the clustroid changes, the new clustroid would be one of them.
  - Point p becomes the new clustroid if $\mathrm{rowsum}(p) < \mathrm{rowsum}(c)$.
- Why the k points farthest from the clustroid?
  - Used to determine whether two clusters are close enough to merge.

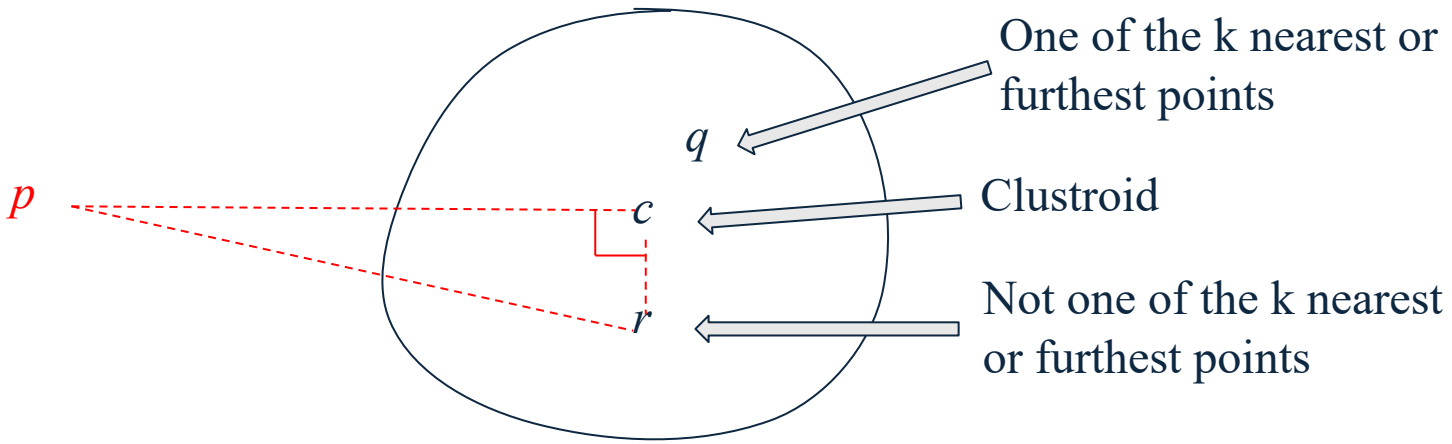
Adding a Point to a Cluster

- How can we add a point p to a cluster?
  - Add 1 to N.
  - For each point $q \in \{\text{clustroid}\} \cup k$ nearest points $\cup\ k$ farthest points, update $\text{rowsum}(q)$ as $\text{rowsum}(q) + d^2(p, q)$.
- What if p needs to be included in the representation?
  - We need to compute the rowsum of p, which needs a disk access.



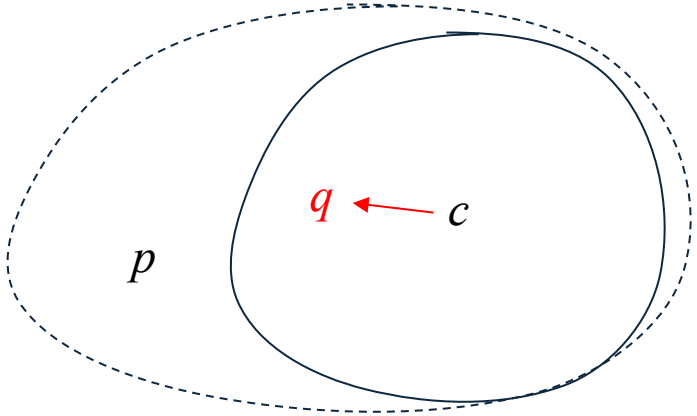
Estimating the Rowsum

- **Proposition:** $\text{rowsum}(p) \approx \text{rowsum}(c) + N \times d^2(p, c)$.
- With the curse of dimensionality, almost all angles are 90°.
- Thus, $d^2(p, r) \approx d^2(p, c) + d^2(c, r)$ by the Pythagorean theorem.



Possibly Updating the Clustroid

- If $\mathrm{rowsum}(q) < \mathrm{rowsum}(c)$ for any q, make it the new clustroid.
- Eventually, the true clustroid may not be in the k closest points.
  - Cluster representation needs to be recomputed periodically from disk.



Other Details of GRGPF

- See the textbook for other details:
  - How to initialize the cluster tree.
  - How to use the tree for each new point.
  - How to split a cluster.
  - How to merge clusters (with the k furthest points).

Summary

1. CURE Algorithm
2. BFR Algorithm
   - Cluster representation
   - Three classes of sets
3. BFR Algorithm: Process
4. GRGPF Algorithm
   - Cluster representation
   - Estimation of a rowsum